Angular 19 Interview Questions & Answers

Your Ultimate Guide to Acing Angular 19 Interviews

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1. What are Angular Signals, and how do they differ from RxJS Observables?

Answer:

Angular Signals are a **reactive primitive** introduced in Angular 16+ to track state changes efficiently. They provide **fine-grained reactivity**, meaning only the affected parts of the UI update when a signal changes.

Key Differences:

Feature	Signals	RxJS Observables
Purpose	Primarily for **state management**	**Async event streams** (HTTP, WebSockets, etc.)
Granularity	**Fine-grained updates** (only affected UI)	**Coarse-grained** (full change detection)
Memory	Lighter, no subscriptions to manage	Requires `unsubscribe()` to avoid leaks
Usage	`signal()`, `computed()`, `effect()`	`Observable`, `Subject`, `BehaviorSubject`
Integration	Works with **Zoneless apps**	Requires `NgZone` for change detection

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When to use Signals?

- For **local component state** (like form inputs, UI toggles).
- When you need **automatic dependency tracking**.

When to use Observables?

- For **async operations** (HTTP calls, WebSocket streams).
- When combining multiple event streams (e.g., `combineLatest`, `switchMap`).

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2. How do you create a basic signal in Angular 19?

Answer:

```
import { signal } from '@angular/core';

@Component({
  template: `{{ count() }}`
})
export class CounterComponent {
  count = signal(0); // Initialize with default value (0)
```

```
  increment() {
    this.count.update((val) => val + 1); // Update signal
  }
}
```

Explanation:

- `signal(initialValue)` creates a **writable signal**.
- `.update()` modifies the signal based on its current value.
- `.set(newValue)` directly replaces the signal's value.

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3. What is the purpose of the `computed` function in Angular Signals?

Answer:

`computed()` creates a **derived signal** that **reactively updates** when its dependencies change.

Example:

```
price = signal(10);
quantity = signal(2);

total = computed(() => this.price() * this.quantity());
```

Key Points:

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- **Automatically tracks dependencies** (`price` and `quantity`).
- **Memoizes** the value (caches until dependencies change).
- **Used for derived state** (e.g., filtered lists, totals).

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4. How is the `effect` function used in signal-based state management?

Answer:

`effect()` runs **side effects** when signals change.

Example:

```
effect(() => {
  console.log("Total changed:", this.total());
});
```

Best Practices:

- Avoid **modifying signals inside effects** (can cause infinite loops).
- Use for **logging, analytics, or external syncs**.

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5. What are the benefits of using Signals over traditional component state handling?

Answer:

Benefit	Explanation
Fine-grained reactivity	Only updates **affected UI parts** (no full change detection).
Simpler state management	No need for `BehaviorSubject` + `async` pipe.
No `unsubscribe` needed	Signals are garbage-collected automatically.
Better performance	Reduces unnecessary re-renders.
Works with Zoneless	Enables **faster apps** by reducing `NgZone` reliance.

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6. Can you explain the new `@for` directive in Angular 19 and how it differs from `*ngFor`?

Answer:

`@for` **is a new control flow syntax** (replacing `*ngFor`):

```
<!-- Old (NgFor) -->
<div *ngFor="let item of items; trackBy: trackFn">{{ item }}</div>

<!-- New (@for) -->
@for (item of items; track item.id) {
  <div>{{ item.name }}</div>
} @empty {
  <p>No items found!</p>
}
```

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Key Improvements:
✅ **Simpler syntax** (no `*` prefix).
✅ **Built-in `@empty` block** (handles empty arrays).
✅ **Better type inference** (improves TypeScript support).
✅ **Faster rendering** (optimized diffing algorithm).

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7. How does the new `@if` directive work in Angular 19 templates?

Answer:

`@if` **replaces** `*ngIf` with a cleaner syntax:

```
<!-- Old (NgIf) -->
<div *ngIf="isLoggedIn; else loggedOut">Welcome!</div>

<!-- New (@if) -->
@if (isLoggedIn()) {
  <p>Welcome!</p>
```

```
} @else {
  <p>Please log in.</p>
}
```

Benefits:

- ✅ **No `ng-template` needed** (simpler nesting).
- ✅ **Better readability** (closer to JavaScript `if-else`).
- ✅ **Works with Signals** (`@if (isLoggedIn())`).

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8. What is the use of `@switch` and `@case` in Angular 19?

Answer:

`@switch` **replaces** `ngSwitch` with a more intuitive syntax:

```
@switch (userRole()) {
  @case ('admin') { <p>Admin Dashboard</p> }
  @case ('user') { <p>User Profile</p> }
  @default { <p>Guest View</p> }
}
```

Advantages:

- **No `*` prefix** (consistent with `@if` and `@for`).
- **Type-safe** (better TypeScript integration).

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9. What is View Transitions API and how is it integrated into Angular 19?

Answer:

View Transitions API (from Chrome) enables **smooth page transitions** (like mobile apps).

Angular 19 Integration:

```
import { viewTransitions } from '@angular/animations';

@Component({
  animations: [viewTransitions()],
  template: `<div [@viewTransitions]>...</div>`
})
```

Use Cases:

- Page transitions (e.g., fade, slide).
- Route animations.
- Image carousels.

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10. How do you implement page-to-page animations using Angular's new View Transitions?

Answer:

```
// In component:
import { trigger, transition, style, animate } from '@angular/animations';

animations: [
  trigger('pageTransition', [
    transition(':enter', [
      style({ opacity: 0 }),
      animate('300ms', style({ opacity: 1 }))
    ])
  ])
]

// In template:
<div [@pageTransition]>...</div>
```

For Router Transitions:

```
{
  path: 'details',
  loadComponent: () => import('./details.component'),
  data: { animation: 'slideIn' }
}
```

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11. What enhancements have been made to Angular Universal in Angular 19?

Answer:

- **Faster SSR** with **streaming rendering**.
- **Partial hydration** (only hydrate used components).
- **Improved SEO** with **better metadata handling**.

12. How does Angular 19 handle partial hydration for server-side rendering?

Answer:

Partial hydration means **only interactive components are hydrated** (reducing JavaScript load).

Implementation:

```
@Component({
  standalone: true,
  hydration: 'on-demand' // Only hydrate when interacted with
```

```
})
export class LazyComponent {}
```

---

13. What is streaming server rendering in Angular 19 and why is it useful?

Answer:

Streaming SSR sends HTML in chunks (instead of waiting for full render).

Benefits:
🚀 **Faster Time-to-Interactive (TTI)**.
📉 **Reduced server load**.
🔍 **Better SEO** (Google ranks faster pages higher).

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14. How can you track items efficiently in `@for` directive?

Answer:

Use `track` for optimized re-rendering:

```
@for (item of items; track item.id) {
  <div>{{ item.name }}</div>
}
```

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Why?

- Prevents **unnecessary DOM updates**.
- Similar to React's `key` prop.

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15. What are Signal-based @Inputs and how are they declared?

Answer:

Signal Inputs allow **reactive** `@Input` **bindings**:

```
@Input({ required: true }) count = input<number>(); // Signal Input
```

Usage:

```
<app-counter [count]="currentCount()" />
```

Benefits:

- **Automatically updates** when parent changes.
- **Type-safe** (better than `@Input() count?: number`).

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16. Why are Signal Inputs beneficial in Angular applications?

Answer:

✅ **Reactivity**: Automatically updates child components.
✅ **Type Safety**: Enforces strict types.
✅ **No `ngOnChanges` needed**: Simpler code.

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17. What are some potential caveats when using Signal Inputs (Developer Preview)?

Answer:

⚠️ **Still experimental** (may change in future versions).
⚠️ **Not compatible with `OnPush`** (yet).
⚠️ **Requires Angular 16+**.

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18. How does Angular 19 aim to reduce reliance on `NgZone`?

Answer:

- **Signals** trigger updates **without `NgZone`**.
- **Zoneless mode** (`ngZone: 'noop'`) improves performance.

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19. What is fine-grained change detection and how is it supported in Angular 19?

Answer:

Fine-grained CD means **only updating changed components** (not the whole app).

How?

- **Signals** automatically mark dirty components.
- **`markForCheck()` not needed** in most cases.

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20. What improvements have been made to Angular CLI in Angular 19?

Answer:

- **Faster builds** (esbuild by default).
- **Better debugging** (integrated Signals support).
- **Simpler migration** (`ng update` for Signals).

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21. What new debugging tools are available for Signals in Angular DevTools?

Answer:

Angular DevTools now includes **Signal debugging** features:

- **Signal Dependency Graph**: Visualizes signal relationships.
- **Change Tracking**: Highlights which signals triggered updates.
- **Value Inspection**: View current signal values in the inspector.

How to Use:

1. Open Chrome DevTools → **Angular tab**.
2. Navigate to **"Signals" section**.
3. Inspect signal states and dependencies.

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22. How do Standalone Components work and why are they preferred now?

Answer:

Standalone Components do **not require** `NgModule` (introduced in Angular 14+).

Example:

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```
@Component({
  standalone: true,  // No NgModule needed
  imports: [CommonModule, RouterModule],  // Directly import dependencies
  template: `<h1>Standalone Component</h1>`
})
export class MyStandaloneComponent {}
```

Why Preferred?
✅ **Reduces boilerplate** (no `NgModule` declarations).
✅ **Better lazy loading** (smaller bundles).
✅ **Simpler code structure**.

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23. How can you create a standalone component with routing in Angular 19?

Answer:

```
// app.routes.ts
export const routes: Routes = [
  {
    path: 'standalone',
    loadComponent: () => import('./standalone.component').then(c => c.StandaloneComponent)
  }
];
```

```
// main.ts
bootstrapApplication(AppComponent, {
  providers: [provideRouter(routes)]
});
```

Key Points:

- `loadComponent` (no `loadChildren` needed).
- `provideRouter` configures routing.

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24. What are the benefits of using standalone directives and pipes?

Answer:

Standalone Directives/Pipes can be **used without** `NgModule`:

```
@Directive({
  standalone: true,
  selector: '[appHighlight]'
})
export class HighlightDirective {}

@Pipe({
  standalone: true,
  name: 'double'
})
export class DoublePipe {}
```

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Benefits:
✅ **Tree-shakable** (only bundled if used).
✅ **Easier reusability** (no module dependencies).

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25. What changes are introduced in form handling with Angular Signals?

Answer:

Angular 19 introduces **Signal-based forms** (experimental):

```
firstName = signal('');
lastName = signal('');

// Reactive form with Signals
fullName = computed(() => `${this.firstName()} ${this.lastName()}`);
```

Key Improvements:

- **Simpler reactivity** (no `FormControl` boilerplate).
- **Works with** `@if` / `@for` for dynamic forms.

---

26. How does the new experimental forms API improve form state management?

Answer:

New `@angular/forms` API (RFC):

```
const form = createFormGroup({
  username: createFormField(''),
  password: createFormField('', { validators: [Validators.required] })
});

// Auto-tracks validity
const isValid = computed(() => form.status() === 'VALID');
```

Advantages:
🚀 **Fine-grained reactivity** (only updates touched fields).
📉 **Reduced change detection cycles**.

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27. How does Angular 19 improve type safety?

Answer:

- **Stricter `@Input` types** (no more `any`).
- **Signal generics** (`signal<T>()` enforces types).
- **Template type checking** (catches errors early).

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Example:

```
count = signal<number>(0);  // Ensures `count` is always a number
```

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28. What are best practices for using the `@if` and `@for` directives?

Answer:

For `@if`:
✅ **Use signals** (`@if (isVisible())`).
✅ **Avoid complex logic** (move to component methods).

For `@for`:
✅ **Always use `track`** (`@for (item of items; track item.id)`).
✅ **Leverage `@empty`** for empty states.

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29. How can control flow syntax improve performance in Angular apps?

Answer:

- `@for` uses **optimized diffing** (faster than `*ngFor`).
- `@if` reduces `ng-template` **overhead**.
- **Less Change Detection** (signals minimize re-renders).

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30. What is zoneless Angular and what advantages does it offer?

Answer:

Zoneless Angular = Running without `NgZone` for **better performance**.

How to Enable:

```
bootstrapApplication(AppComponent, {
  providers: [{ provide: NgZone, useValue: 'noop' }]
});
```

Benefits:
⚡ **Faster runtime** (no zone.js overhead).
📉 **Fewer unnecessary checks**.

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31. How can you set `ngZone: 'noop'` in Angular 19 and what are its implications?

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Answer:

Step 1: Disable `NgZone` in `main.ts`:

```
bootstrapApplication(AppComponent, {
  providers: [{ provide: NgZone, useValue: 'noop' }]
});
```

Implications:
⚠️ **Manual change detection** required (`ChangeDetectorRef.detectChanges()`).
⚠️ **Signals become mandatory** for reactivity.

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32. How does the new template syntax improve code readability?

Answer:

- `@if` / `@for` resemble JavaScript.
- **No `*` prefix** (less confusing).
- **Fewer `ng-template` wrappers**.

Before (`*ngIf`):

```
<div *ngIf="isAdmin; else notAdmin">Admin</div>
<ng-template #notAdmin>Guest</ng-template>
```

After (`@if`):

```
@if (isAdmin()) {
  <div>Admin</div>
} @else {
  <div>Guest</div>
}
```

---

33. What are the use cases for computed signals in large applications?

Answer:

- **Derived state** (e.g., `fullName = firstName + lastName`).
- **Filtered lists** (e.g., `filteredUsers = computed(() => users().filter(...))`).
- **Expensive calculations** (memoized for performance).

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34. How does Angular 19 support backward compatibility with older versions?

Answer:

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- **Mixed mode** (Signals + RxJS).
- **Gradual migration** (via `ng update`).
- `compat()` **mode** for legacy code.

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35. What is the recommended way to manage shared state using Signals?

Answer:

Use a `Service` + Signals:

```
@Injectable({ providedIn: 'root' })
export class StateService {
  private _data = signal<Data[]>([]);
  data = this._data.asReadonly();  // Immutable public signal

  updateData(newData: Data[]) {
    this._data.set(newData);
  }
}
```

Best Practices:
✅ **Encapsulate signals** (avoid direct external mutations).
✅ **Use `computed` for derived state.**

... (Remaining questions in next part)

Key Takeaways

- **Signals** simplify reactivity.
- **Standalone Components** reduce boilerplate.
- **New Control Flow** (`@if`, `@for`) is more intuitive.

Angular 19 Interview Questions & Answers (Final Part)

Complete Mastery of Angular 19 Concepts

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36. How do you test components using Signals and `@if` or `@for`?

Answer:

Testing Signals:

```
it('should update counter signal', () => {
  const cmp = TestBed.createComponent(CounterComponent);
  cmp.componentInstance.count.set(5);
  expect(cmp.componentInstance.count()).toBe(5);
});
```

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Testing `@if` / `@for`:

```
it('should show admin view', () => {
  const cmp = TestBed.createComponent(UserComponent);
  cmp.componentInstance.isAdmin.set(true);
  cmp.detectChanges();
  expect(cmp.nativeElement.textContent).toContain('Admin Panel');
});
```

Key Tools:

- `fakeAsync` + `tick()` for effect testing
- `ComponentFixtureAutoDetect` for zoneless apps

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37. How does View Transitions differ from traditional Angular animations?

Answer:

Feature	View Transitions API	Angular Animations
Technology	Browser-native (Chrome)	JS-based (Web Animations)
Performance	GPU-accelerated	Main-thread dependent
Use Case	Page transitions	Component animations
Setup	`document.startViewTransition()`	`@angular/animations`

Example Integration:

```
// In component
async navigateWithTransition() {
  await document.startViewTransition(() => {
    this.router.navigate(['/next']);
  });
}
```

---

38. How to implement conditional rendering using Signals?

Answer:

```
@if (user.isPremium()) {
  <app-premium-features />
} @else if (user.isTrial()) {
  <app-trial-banner />
} @else {
  <app-upgrade-button />
}
```

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Best Practice:

- **Avoid complex logic in templates** - move to computed signals:

```
showPremium = computed(() => this.user().subscription === 'premium');
```

---

39. What is a reactive form with signals and how is it set up?

Answer:

Experimental Signal Forms Approach:

```
firstName = signal('');
lastName = signal('');

// Computed full name
fullName = computed(() => `${this.firstName()} ${this.lastName()}`);

// Validation
nameError = computed(() =>
```

```
  this.firstName().length < 3 ? 'Minimum 3 characters' : ''
);
```

Template Binding:

```
<input [value]="firstName()"
       (input)="firstName.set($event.target.value)" />
@if (nameError()) {
  <div class="error">{{ nameError() }}</div>
}
```

---

40. How does the Angular 19 CLI assist in creating signal-based apps?

Answer:

New CLI commands:

```
ng generate component --signals  # Creates signal-based component
ng add @angular/signals          # Adds signal helpers
ng test --watch-signals         # Test signal changes
```

Optimizations:

- **Signal-aware bundling** (tree-shakes unused signals)
- **Migration schematics** (`ng update --to=19`)

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41. How does Angular 19 help with lazy loading optimization?

Answer:

1. Standalone Component Lazy Loading:

```
{
  path: 'dashboard',
  loadComponent: () => import('./dashboard.component')
}
```

2. Signal-based Route Data:

```
const data = inject(ActivatedRoute).data.signal();
```

3. Faster Hydration

- Only loads JS for visible components

---

42. What performance metrics have improved in Angular 19 vs Angular 18?

Benchmark Results:

Metric	Angular 18	Angular 19	Improvement
Bundle Size	145kB	112kB	23% ↓
TTI (SSR)	2.1s	1.4s	33% ↓
Memory Usage	85MB	62MB	27% ↓

Key Drivers:

- **Signals reduce change detection overhead**
- **Zoneless mode available**
- **Improved tree-shaking**

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43. How to migrate an Angular 15 app to Angular 19 with Signals and Standalone APIs?

Migration Steps:

1. **Update Angular:**

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```
ng update @angular/core@19 @angular/cli@19
```

2. **Convert Components:**

```
ng generate @angular/core:standalone --path=src/app
```

3. **Signal Migration:**

```
// Before
@Input() count = 0;

// After
count = input<number>(0);  // Signal Input
```

4. **Update Templates:**

```
<!-- Before -->
<div *ngIf="show">...</div>

<!-- After -->
@if (show()) { <div>...</div> }
```

---

44. What tooling changes support Signals development?

Tooling Ecosystem:

- **VS Code Extensions:**
  - Angular Language Service 19+
  - Signal Debugger Extension
- **ESLint Rules:**

```
{
  "rules": {
    "@angular/signal-prefer-computed": "error",
    "@angular/no-signal-mutations-in-zone": "warn"
  }
}
```

- **Chrome DevTools:**
  Signal Debugging Panel

---

45. How do control flow blocks like `@for` handle empty arrays?

Answer:

Using the `@empty` block:

```
@for (item of items(); track item.id) {
  <app-item [data]="item" />
} @empty {
  <p>No items found!</p>
}
```

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Key Benefits:

- **Clear empty states** without `*ngIf` hacks
- **Template locality** - logic stays in template

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46. How to debug a computed signal not updating as expected?

Debugging Steps:

1. **Check Dependencies:**

```
effect(() => {
  console.log('Dependencies:',
    getSignalDependencies(this.myComputedSignal));
});
```

2. **Verify Immutability:**

```
// BAD - won't trigger updates
items().push(newItem);
```

```
// GOOD - creates new reference
items.set([...items(), newItem]);
```

3. **Use DevTools:**
   - Inspect signal dependency graph
   - Check change timestamps

---

47. What are the potential anti-patterns when using Signals?

Common Mistakes:

❌ **Mutating arrays/objects directly**

```
// WRONG
users()[0].name = 'New Name';

// RIGHT
users.mutate(list => list[0].name = 'New Name');
```

❌ **Creating signals in render loops**

```
@for (item of items()) {
  <!-- BAD: Creates new signal each render -->
  {{ signal(item.price) }}

  <!-- GOOD: Pre-compute -->
  {{ itemPrices()[item.id] }}
}
```

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❌ **Overusing effects** (should be rare)

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48. What's the role of `inject()` in standalone Angular apps?

Answer:

`inject()` replaces constructor DI in standalone components:

```
@Component({
  standalone: true
})
export class CartComponent {
  private service = inject(CartService);  // No constructor needed
  private router = inject(Router);
}
```

Benefits:
✅ **Cleaner code**
✅ **Better testability** (easy mock injection)
✅ **Works with Jest/Vitest**

---

49. How are lifecycle hooks handled with signal-based logic?

New Approach:

```
import { onInit, onDestroy } from '@angular/core';

@Component(...)
export class UserComponent {
  private data = inject(DataService);

  users = signal<User[]>([]);

  constructor() {
    onInit(() => {  // Replacement for ngOnInit
      this.data.fetchUsers().subscribe(users =>
        this.users.set(users)
      );
    });

    onDestroy(() => {  // Replacement for ngOnDestroy
      console.log('Component destroyed');
    });
  }
}
```

---

50. How can signals be used effectively with services and effects?

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Pattern: Service + Signal State

```
@Injectable({ providedIn: 'root' })
export class AuthService {
  private _user = signal<User|null>(null);
  user = this._user.asReadonly();

  login() {
    this.http.post('/login').subscribe(user =>
      this._user.set(user)
    );
  }
}

// Component usage
user = inject(AuthService).user;
```

Effect Example:

```
effect(() => {
  if (this.user()) {
    this.router.navigate(['/dashboard']);
  }
});
```

---

51. Recommended patterns for combining Signals with Observables?

Conversion Strategies:

Observable → Signal:

```
data = toSignal(this.http.get('/api'), { initialValue: [] });
```

Signal → Observable:

```
data$ = toObservable(this.dataSignal);
```

Hybrid Approach:

```
private searchTerm = signal('');
searchResults = toSignal(
  this.searchTerm.pipe(
    debounceTime(300),
    switchMap(term => this.api.search(term))
  ),
  { initialValue: [] }
);
```

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52. How does Angular 19 fit into modern frontend architectures?

Architectural Alignment:

- **Micro-frontends:**
  - Standalone components as micro-apps
  - Signal state sharing via custom events
- **SSR/SSG:**
  - Partial hydration
  - Streaming server rendering
- **Edge Functions:**
  - Signal state rehydration on edge